

SHIVA KARTHIK KOILAKONDA

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Education

MVSR Engineering College

B.Tech in Electronics and Communication Engineering GPA: 8.64 / 10

Nov 2022 – May 2026

Hyderabad, India

Gouthami Junior College

Intermediate, MPC Percentage: 98.2 / 100

Jun 2020 – Mar 2022

Nalgonda, India

SPR High School

School CGPA: 10.0 / 10

Jun 2008 – Mar 2020

Nalgonda, India

Experience

Crystallite Technologies

Frontend Development Intern

Mar 2025 – Sep 2025

Remote

Projects

Automatic Braking System | Arduino, Ultrasonic Sensor, Accelerometer, Servo Motor, Relay

- Developed an automatic braking system to enhance vehicular safety by detecting obstacles and applying brakes using a servo motor.
- Utilized an ultrasonic sensor to measure the distance between the vehicle and obstacles, triggering braking action when a collision risk is detected.
- Integrated an accelerometer to detect motion and simulate real-world braking scenarios.
- Created a real-time feedback system displayed on a 16x2 LCD.
- Software used: Arduino IDE.

Fake Currency Detection System using Neural Networks | Neural Networks, Image Sensor, Python, FPGA

- Developed a real-time system to detect counterfeit Indian currency notes, improving financial security.
- Captured high-resolution images under visible light and UV illumination for model training.
- Implemented MobileNetV2, EfficientNet, and ResNet models for accurate classification with hardware efficiency.
- Prototyped on an FPGA platform using Xilinx Vitis AI and High-Level Synthesis (HLS) for real-time verification.
- Demonstrated fast and reliable detection suitable for banks, businesses, and daily transactions.

Technical Skills

Core Areas: Digital Logic Design, Microcontrollers, Embedded Systems, Analog & Digital Communication

Programming Languages: C, Python, MATLAB

Simulation & Design Tools: MATLAB, Proteus, Xilinx Vivado

Hardware Platforms: Arduino, 8051 Microcontroller

Soft Skills: Problem-Solving, Team Collaboration, Analytical Thinking, Attention to Detail

Certifications and Achievements

Certified in Python Programming – ElSystems

Certified in Java – Spoken Tutorial Project, IIT Bombay

Personal Interests / Hobbies

Exploring electronics and building circuits

Passionate about outdoor sports

Traveling and discovering new cultures

Cooking and exploring new cuisines